

Remote sensing and high-frequency monitoring

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9.1 Introduction

Remote sensing measurement for water quality monitoring is known for 50 years and has been developed with new satellites and improved onboard sensors. Since then, a lot of works have been published, among which a recent review by Chawla et al. [1]. In the same time, new handheld or online devices were proposed for a better knowledge of water quality as the fine structure of water quality dynamics [2]. A few years ago, Rode et al. presented in their review [3] the advancement of technologies for in situ water quality measurement with some important applications. However, some recent works as the one by O'Grady et al. [4] linked both approaches (remote and on-site) within a tiered framework of integrated sensing technologies. They assume that the future of monitoring will involve satellite, in situ and airborne devices with data analytics playing a key role in providing decision support tools.

However, monitoring applications are still grouped either under remote sensing from satellites or under on-site high-frequency measurements, carried out under different ways.

Fig. 9.1 represents the different levels of water monitoring, from airborne systems (satellites, aircrafts, and drones) to underwater ones, with floating devices (boats and buoys) and wireless sensor network (WSN) if any. The scene of monitoring is completed by in situ or online measurements of water quality.

This chapter is organized from the top-down gradients of Fig. 9.1, from satellites applications to underwater observations, and considers

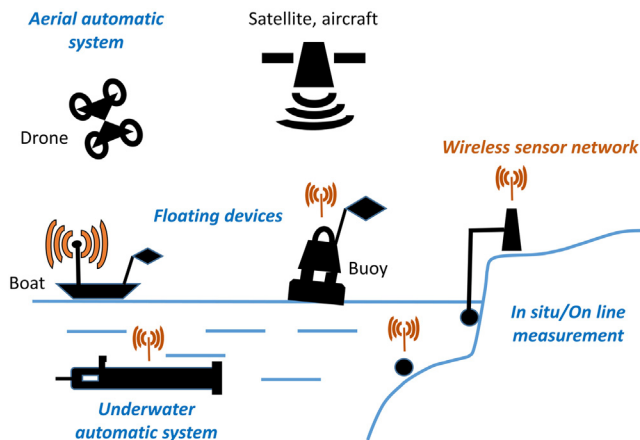


FIGURE 9.1 Remote and high-frequency systems for water quality monitoring. *Credit: O. Thomas, M-F. Thomas and J. Causse.*

only works carried out with optical sensors, which are the great majority of remote sensing at large applications.

9.2 Satellites applications

Water has specific reflectance characteristics, which can be measured at different wavelengths, based on the scattering response of suspended solids and on the absorbing properties of some of its dissolved components [chlorophyll, colored dissolved organic matter (DOM) (CDOM), etc.]. With the knowledge of their optical characteristics, it is possible to retrieve quantitative values of the concentrations for these water constituents, solely based on the reflectance of light measured by satellite sensors. Thus selective sensors boarded on satellites can remotely carry out the measurement of radiation reflected by the Earth's components, including water surface. Remote sensing measurement for water quality monitoring is known for 50 years. It was first proposed with the observation satellites launching around the 1970s. In a recent review, Topp et al. [5] showed that a huge number of scientific and technical papers have demonstrated the interest of remote sensing for the estimation of biological, chemical, and physical properties of inland water bodies and seawater in addition to soil, territory, and vegetation data. This review showed the progression of inland water quality remote sensing from methodological development to scientific applications. At the beginning, the majority of publications focused on model development and validation followed by a spike in publications, beginning in the early 2000s, applying remote sensing models to analyze spatiotemporal trends, drivers, and impacts of changing water quality on ecosystems and human populations. Works in the past decade showed that researchers are using improved computing resources, datasets, and operational remote sensing algorithms to better understand complex inland water systems. Future satellite missions promise to continue these improvements by providing observational continuity with spatial/spectral resolutions ideal for inland waters [5].

In a recent study, Du et al. [6] reported the evolution of total suspended matter (TSM) concentrations in waters using Landsat images during almost two decades across the Songnen Plain, China. From 2012 to 2015, a total of 142 water samples were collected from 22 water bodies for the model validation. The sampling point number for each water body was determined according to its water area with a minimum distance of 500 m observed between points.

Different sensors were available according to Landsat flights: TM/Thematic Mapper instruments for Landsat 5, ETM/Enhanced Thematic Mapper Plus scanner for Landsat 7, and OLI/Operational Land Imager

for Landsat 8. Images without clouds or snow from April to October were only selected from 1984 to 2018.

This study demonstrated the potential of using historical Landsat imagery combined with field survey data to map the spatiotemporal patterns of TSM in inland waters. A model was established for TSM concentration between band combinations from Landsat' sensor images and field-measured TSM concentrations. The results showed that TSM concentrations in most of waters were decreasing gradually from 1984 to 2018 and that of all water bodies generally decreased from April to the minimum level in July and then increase from September. Moreover, high spatial heterogeneities of the TSM across the study area were found with TSM in reservoirs generally lower than in lakes. TSM concentrations in the larger lakes were usually lower than those in the smaller lakes.

Finally, the study also revealed the potential role of wind speed and accumulated precipitation in regulating TSM changes. In addition, increasing of vegetation can strongly reduce the TSM in waters.

Among sensors used for remote measurement of water quality, MODIS (Moderate Resolution Imaging Spectroradiometer) is one major instrument often chosen for the understanding of global dynamics and processes occurring on the land, in the oceans, and in the lower atmosphere. MODIS is viewing from 2000 the entire Earth's surface every 1 or 2 days. A recent study published by Feng et al. [7] used long-term MODIS observations for the monitoring of the water transparency of 50 large lakes on the Yangtze Plain (China). A semianalytical algorithm for water clarity estimation proposed by Lee et al. [8] was used for water transparency calculation from MODIS observations between 2003 and 2016, expressed as the Secchi disk depth (Z_{sd}) estimation.

The results showed significant seasonal and interannual changes in water clarity and differences between lakes, with human activities influencing more lakes than natural variability. The Z_{sd} results showed different changing patterns from those described in previous turbidity studies, as water turbidity only reflects concentrations of total suspended solids (TSS), while the Z_{sd} also takes into account the compound effects of optically active substances (chlorophyll-a, TSS, and CDOM). This study highlights the importance of satellite observations and robust algorithms in obtaining large-scale water quality information.

A multidecade comparison of remote sensing data was proposed by Carstens and Amer [9] who showed changes in urban development and water quality of the Pontchartrín Basin (Louisiana) within 30 years. A Landsat Thematic Mapper image from 1985 and a Landsat Operational Land Imager from 2015 were processed and compared. Additional in situ water quality data were considered. The results indicated that urbanization has a negative impact on water quality.

The development of impervious surfaces and related urbanization area has led to increasing water pollution with some water quality parameters higher (fecal coliform, phosphorus) and some lower (dissolved oxygen).

A recent report by Ledang et al. [10] presented the results of tests on the application of remote sensing data for selected lakes included in the lake monitoring programs run by the Norwegian Environment Agency. Monitoring of lake water quality is challenging in Norway due to the large number of lakes, and their remote location. The data of three satellites were considered (1) Envisat, launched in 2002 with the MERIS sensor (Medium Resolution Imaging Spectrophotometer) operating between 390 and 1040 nm. Its revisiting time for Norway is 2–3 days and its resolution is around 300 m on the Earth. (2) Sentinel-2 launched in 2015 in the frame of program Copernicus, with a multi-spectral instrument (MSI) working at 13 bands and giving a resolution of 60 m for water bodies. Like Envisat, it is available for Norway every 2–3 days. (3) Sentinel-3 launched for sea surface, temperature, and color observation boarding four main instruments, including the Ocean and Land Color Instrument based on the MERIS design. Its spatial resolution was 300 m. The data acquired by these three satellites were used for the estimation of some water quality parameters (chlorophyll-a, TSM, and CDOM) after atmospheric correction of sensors signals.

Five different lakes have been studied in this work. Data from MERIS, Sentinel-2, and Sentinel-3 showed general good results for chlorophyll-a, TSM, Secchi disc depth (z_{SD}), and turbidity compared to *in situ* measurements. However, an overestimation of chlorophyll-a was observed when the reflectance signal of water is very low. In this case, some ways of signal improvement should be searched (better atmospheric compensation, better estimation tools).

In a recent paper by Erena et al. [11], the water quality of a 20 km length coastal lagoon (Mar Menor, Southern-East of Spain) was investigated between May 2015 and October 2017. Remote sensing data from Landsat 8 and Spot 7 and four different sensors (TM, ETM, OLI, and MERIS) were acquired with complementary weekly field sample analysis and measurements of chlorophyll-a, phycocyanin (for the assessment of algae blooms), and turbidity. Spatial remote sensing has detected that turbidity in the Mar Menor is not caused only by phytoplankton; there are also chemical compounds involved, interacting with phytoplankton and leading to changes in color.

Besides the abovementioned works for inland water bodies, the observation of seawater quality has given numerous applications that are presented in Table 9.1.

TABLE 9.1 Some applications of remote sensing from satellites for seawater quality monitoring.

Location	Date	Satellite/sensor ^a	Parameters	References
Estuarine waters, Australia	1990–91	Landsat/TM	Chlorophyll-a, transparency (Z_{sd}) salinity	[12]
Gulf of Finland	1997	Landsat/TM, ERS2/SARdata	Turbidity, transparency (Z_{sd}), suspended solids	[13]
Gulf of Gabes, Tunisia	2009	Aqua/MODIS	Chlorophyll-a, Turbidity	[14]
Baltic Sea	2008–10	Envisat/MERIS	Chlorophyll-a	[15]
Baltic Sea	2016	Sentinel3A/OLCI	Chlorophyll-a, total suspended solids, colored dissolved organic matter	[16]
North Sea	1994–2017	Various/Modis, MSI, OLI	Seagrass (<i>Zostera</i>)	[17]
South China Sea	1979–2014	Various/CZCS/SeaWiFS, MODIS	Chlorophyll-a	[18]
Different oceanic waters	1997–2013	Various/MODIS, MERIS, SeaWiFS	Color (Forel-Ule)	[19]

^aTM: Thematic Mapper/ERS2/SARdata: Synthetic Aperture Radar data/MODIS: Moderate Resolution Imaging Spectroradiometer/MERIS: Medium Resolution Imaging Spectrometer/OLCI: Ocean Land Colour Instrument /MSI: Multispectral Instrument/ OLI: Operational Land Imager/CZCS: Coastal Zone Color Scanner/SeaWiFS: Sea viewing Wide Field of view Sensor.

9.3 Other airborne applications

9.3.1 Aircrafts

Considering the strong development of satellite remote sensing, aircrafts were rather few used except for military imaging applications. However, a first study reported by Grenier et al. [20] described a project for the study of water quality of Saint-Laurent stream (Quebec), downstream Montreal, by using data from multispectral optical sensors (MEIS II, PMI) boarded in aircrafts flying at 12,000 m height and giving a 4 m ground resolution. Simultaneous water samples were taken with aircraft passages in October 1989, for the laboratory measurement: turbidity, color, dissolved organic carbon (DOC), chlorophyll, and TSS. After calibration, the regression analysis gave determination coefficients values between 0.6 and 0.9 depending on the sensor and spectral bands used.

In the early 2000s, a review by Ritchie et al. [21] cited some remote sensing applications by spectral sensors on aircraft to study sediment patterns from reflectance measured between 700 and 800 nm and distribution of chlorophyll in Chesapeake Bay from reflectance around 550 and 700 nm [22]. The Ocean Data Acquisition System (ODAS) instrument used was composed of three radiometers with 15 nm bands and the aircraft flew at around 500 m giving a spatial resolution of 5 m.

9.3.2 Drones

In his review, McDonald [23] reported that drones or unmanned aerial vehicles (UAV) have applications for urban stormwater management. More than 10 works were cited for applications on rivers, reservoirs, or lakes water quality. All UAV were equipped with camera or multispectral instruments and have shown good prediction of surface water quality parameters such as turbidity, suspended solids, and chlorophyll. This was possible from the exploitation signal of red, green, and blue (RGB) regions of visible spectrum for camera or from the 400–500, 660, or 780 nm wavelengths for spectral sensors.

A recent work, Kupssinskü et al. [24] described a methods for chlorophyll-a and suspended solids prediction through remote sensing and machine learning. Water samples must be collected for modeling with the remote signal from Sentinel-2 satellite equipped with the MSI or a camera/sensor boarded on a drone. Spectral bands from visible to infrared are available for the purpose from MSI or RGB signal from camera drone. A comparison of both systems showed a better resolution for drone and a higher temporal frequency.

More recently, an innovative research reported the assessment of microbial water quality in an irrigation pond from drone-based imaging [25]. Several cameras with different set of lens were mounted on a drone flying at 400-m height. The resulting images were exploited through regression tree approach, taking into account the characteristics of water samples. This study has shown that UAV imagery provided the same level of accuracy in estimating concentrations of *Escherichia coli* across the study pond, compared to in situ measurements of physico-chemical and biological water quality parameters.

Besides classical optical sensors (camera, spectral sensor) boarded on drones and measuring reflectance or absorbance of surface water, some fluorescence applications can be reported such as the one of Duan et al. [26].

High-quality fluorescence spectra of the superficial aquatic layer featuring spectral signatures from oils' DOM, algae, etc. were acquired by a compact and lightweight laser-induced fluorosensor (LIF), boarded an aerial drone. The system provided a good alternative to conventional pulsed-laser fluorosensors and continuous wavelength (CW)-laser

Scheimpflug systems and exhibited certain advantages in terms of performance and simplicity. Even if it could only be used at nighttime, several experiments were carried out on oil spills in a river showing a strong fluorescence signal at 500 nm. Using sufficiently high CW powers and combining with a suitable modulation scheme for background subtraction, daylight operation should be feasible, considerably extending the present capability. This drone-based remote LIF technique was shown to be a relevant detection technique, especially in the studies of limited areas.

9.4 Surface applications

9.4.1 Boats and buoys

Even if the solution seems to be evident, there are few experiments for water quality monitoring from boats. Levin et al. [27] studied the inherent optical properties in the Baltic Sea from reflectance measurements at 550 nm during 14 cruises performed in the southern Baltic in 1999–2005 onboard the research vessel “Oceania.” The results showed that data from the Baltic Sea were close to the ones acquired in other places (Pacific ocean, Arabian sea).

Besides experiments from research vessels, some tentative works were carried out with unmanned remote boats. A first work was carried out for the study of the dilution treated wastewaters in receiving medium [28]. A radio-controlled 4' mini boat was designed for sampling and measurement of physicochemical parameters (temperature, dissolved oxygen, and conductivity). After sampling, the measurement of nitrate, total organic carbon (estimated), and other specific compounds such as phenols is carried out on the bank with a portable UV multi-channel sensor. It was used for the study of urban and industrial treated wastewater discharges near Lyon (France). This device was built up after a first prototype designed for run-of-the river sampling and measurement in a 12 km evacuation pipe of treated wastewater in Chambéry (France) [29]. A simple floating raft of 3" diameter was equipped with an autosampler and a measurement part, including a multiparameter probe, for sampling and measurement at fixed interval.

A nautical drone was designed for the monitoring of estuarine and coastal waters in the context of Water Framework Directive by Daniel et al. [30]. It was able to go up to 500 m off the coastline in less than 5 minutes; take measurements of temperature, salinity, and turbidity; and collect water samples for biological laboratory analysis.

An Autonomous Surface Vehicle for Water Quality Monitoring was designed by Dunbabin et al. [31] in Australia. A 16' solar powered

catamaran was deployed on Lake Wivenhoe with an array of 50 floating sensor network nodes for communication and event detection. Several sensors were integrated: multiparameter probe (measuring temperature, conductivity, chlorophyll, turbidity, dissolved oxygen, and incident radiation); wind sensor; and profiling sonar. Another monitoring boat was proposed by Hitz et al. [32] with the specificity of having a submersible multiparameter probe for measurement from surface to 50' depth. It was used for the study of cyanobacteria blooms in lake Zurich (Switzerland).

Buoys are sometimes used instead of boats or surface vessel. Chaffin et al. [33] studied the accuracy of data buoys for measurement of cyanobacteria, chlorophyll, and turbidity in a large lake (Lake Erie, North America). With the aim of the detection of cyanobacterial blooms from water quality probe measurements, water sampling and measurement were carried out close to two buoys equipped with a multiparameter probe, including fluorescence detection. Significant correlations occurred between buoy data and water sample data for cyanobacterial biomass, total algae, and turbidity. However, probes clogging and interferences may occur leading to the need of regular cleaning and calibration. Moreover, the measurement of nitrate concentration could complete the procedure of early warning for potentially toxic blooms.

Several autonomous buoys can be used for the monitoring of large water bodies [34]. The study reported the implementation of a small fleet of 50 buoys deployed on a lake or reservoir for monitoring objective. Each buoy is a 1-ft-diameter spherical system with a sensory probe, an external motor and propeller, GPS, and remote communication system. The environmental sensors are limited to a multiparameter probe for basic physicochemical parameters, but optical sensing could be added if any.

More recently, Zhou et al. [35] studied flushes of large amounts of organic material into lakes in the event of rainstorm in China. A multiparameter probe with a fluorimetric sensor was fixed on two buoys for the measurement of fluorescent DOM (FDOM). Field samples were collected during storm events in order to complete buoys' information. The results suggest that rainstorm events enhanced the export to the lake of colored, hydrophobic, and aromatic DOM. The application of in situ fluorescence sensors provides an early warning of DOC surge incidents caused by rainstorm events and may be useful in advising drinking water treatment plant managers of changes in raw water DOM quality and treatability.

Finally, Claustre et al. [36] recently described the Biogeochemical-Argo (BGC-Argo) network of profiling floats carrying sensors that enable observation of as many as six essential biogeochemical and biooptical variables: oxygen, nitrate, pH, chlorophyll-a, suspended

particles, and downwelling irradiance, using UV–visible detection for some parameters. Able to move vertically up to 2000-m depth and horizontally, before sending the acquired data from the surface through the Iridium satellite, floats can bring to the community of oceanographic researchers, essential data for important issues such as the ocean acidification or ocean carbon uptake. The extension of this network has the potential to become a pivotal observation system that links satellite and on-site (ship-based, for example) observations in the near future.

9.4.2 High-frequency grab sampling

In their study, Hansen and Singh [37] used high-frequency sensor data to reveal across-scale nitrate dynamics in response to hydrology and biogeochemistry in Intensively Managed Agricultural Basins. The analysis was conducted on 5-year USGS (United States Geological Survey) sensor data from eight riverine locations within Iowa, United States, including nitrate, discharge, and water temperature, acquired every 15 minutes between 2012 and 2017. The results show that nitrate (NO_3) was strongly correlated with discharge for all sites and correlation increased significantly with drainage area and with crop coverage across watersheds. However, the highest NO_3 concentrations reported often occurred at intermediate-sized discharge events, with high-peak NO_3 likely to occur in response to projected regional changes in climate.

In a recent work, Qin et al. [38] used high-frequency monitoring to reveal how hydrochemistry and dissolved carbon respond to rainstorms at a karstic critical zone, Southwestern China. Around 50 grab samples were taken daily for the analysis of DOC and major ions, and water temperature (T), dissolved oxygen (DO), electrical conductivity (EC), pH, and discharge were continuously measured at high temporal resolution (15-minute interval) using a multiparameter probe and a flowmeter. The results showed that the hydrochemical behavior and dissolved carbon dynamics are highly sensitive to hydrological variations in the karstic critical zone influenced by monsoon, which has high chemical transformation rates and experiences strong anthropogenic impact.

High-frequency monitoring is more often applied for the study of nitrate dynamics but other experiments can be cited. In a recent work, Rosset et al. [39] used high-frequency monitoring fluorescence during 3 years for the study of drivers of seasonal and event-scale DOC dynamics at the outlet of two mountainous peatlands. For that, a fluorescence sensor gave every 30 minutes a FDOM signal which was correlated by DOC measurement from grab samples. The DOC concentration time series can be decomposed into a seasonal baseline interrupted by many short, intense peaks of higher concentrations. At the seasonal scale,

DOC concentration baseline variations are mainly explained by peat water temperature, which controls integrative DOC production processes within the peatland. Finally, peatlands should not be considered as uniform landscapes. Distinct peatland units within the same peatland complex contribute differently to the DOC transfer processes to inland waters. In situ fluorescence measurement was also recently studied [40] combined with rainfall radar data, to provide new insights into natural organic matter transport dynamics in an urban river.

Recent application of high-frequency grab sampling was for monitoring of nitrate isotopes (15 and 18) for pollution sources tracking in two watersheds in China [41]. This approach appeared to be more relevant than the traditional monitoring protocol based on measurements conducted during few rainfall events.

9.4.3 Handheld devices

An integrated water quality monitoring device placed in a backpack was designed for the quick measurement of several parameters thanks to a multiparameter probe, and an innovative UV sensor. On the station chosen, the water to be analyzed is pumped into the device using a sampling tube integrating a pump immersed in the watercourse. The multiparameter probe allows the direct measurement of the electrical conductivity, temperature, pH, oxidation–reduction potential, turbidity, and oxygen content of the water sampled. The water is then analyzed using a double optical path UV–visible spectrophotometer adapted to different concentration levels. Nitrate and DOC concentrations can be easily determined [42].

After results displayed on smartphone screen, within 1 minute, the water is discharged. This system was validated during a research project [43] that was conducted for 1 year (September 2018–December 2019) for the identification of nitrate emission and abatement zones at a fine spatial resolution, by measuring water quality every 50 to 100 m in the watersheds of the Bay of Douarnenez and the Lieue de Grève (Brittany, France). The portable measurement was validated from water samples and lab analysis. The adjustment was excellent with a determination coefficient of 0.99 and 0.95 for nitrate and Dissolved Organic Carbon (DOC), respectively (for more than 200 samples). Furthermore, the qualitative analysis of evolution of spectra along the river continuum, including all tributaries, drains, ditches, plant rejects, and other point sources observed in the field is a powerful tool to interpret hydrology and anthropic impacts in watersheds, complementary to all well-known parameters measured. The designed procedure makes it possible to increase the number of subwatersheds that can be

monitored in the same day, and to increase the resolution of the measurement networks from one point per several tens or hundreds of km^2 to one point per subwatershed of a few km^2 . High spatial resolution measurement allows the localization of highly charged groundwater resurgences and the identification of potential sites for the development of buffer zones. In all contexts, measurement at high spatial resolution allows a better understanding of the functioning of a watershed and the nature of the flows in space, and thus time, that structure the watershed. It also allows the definition of strategies for the implementation of agri-environmental measures within green algae control plans and the nature of a temporal monitoring network.

Besides UV–visible spectrophotometry, handheld fluorometer was also proposed by Chen et al. [44]. A field, lightweight laser fluorometer based on the method of laser-induced fluorescence was developed for water quality monitoring. The hyperspectral LIF technique has the potential to simultaneously detect multiple water quality parameters of interest, which is particularly advantageous in optically complex waters. Under the laser excitation at 405 nm, the peak near 685 nm in the emission spectra corresponds to chlorophyll-a, the peak nearby 470 nm corresponds to Raman scattering (also linked to TSM response), and the peak near 508 nm corresponds to CDOM constituent. Simultaneous estimates of chlorophyll-a, CDOM, and TSM measured by the laser fluorometer were observed to agree well with those measured by laboratory methods after sampling, ($R^2 > 0.85$) in Hangzhou Bay water (China). This procedure allowed successful high-resolution and high-frequency monitoring in a complex estuarine system.

9.4.4 On-site systems

In a recent study, Luna Juncal et al. [45] proposed a mobile monitoring station integrated to a trailer, for the real-time measurement of nitrate and physicochemical water quality parameters every 15 minutes. Several experiments were successfully conducted in Australia showing that contrary to traditional high-frequency monitoring systems (not easy to move) [46], the mobile system developed can be deployed at different target locations on demand.

Werner et al. [47] used a UV sensor to acquire UV absorption values from a headwater stream in central Germany, every 15 minutes for more than 1 year. After the DOC calibration by Partial Least Squares Regression (PLSR) from UV signal of 28 grab samples, SUVA₂₅₄ and S_{275–295}, slopes of UV spectrum between 275 and 295 nm, were calculated. The results provide a mechanistic explanation of the seasonally changing characteristics of DOC–discharge relationships and therefore

can be used to infer the spatiotemporal dynamics of DOC origin in riparian zones from the DOC dynamics of headwater streams.

Nitrate or DOC are not the only parameters studied with high-frequency monitoring. Solids fluxes dynamics were also followed by using turbidimetry correlation [48] for TSS assessment, and conductimetry as a proxy for total dissolved solids (TDS). An experiment over a period of 12 months with 15 minutes frequency measurement was conducted in a mixed restored prairie and agricultural catchment in Nebraska. TDS fluxes were about three times higher than TSS fluxes from the catchment as a whole. However, the TSS fluxes were higher in the agricultural section of the catchment than from the restored prairie.

Mistick and Johnson [49] recently studied high-frequency analysis of DOC storm by UV sensor correlation, the responses in headwater streams of contrasting forest harvest history. Two headwater streams were selected for continuous monitoring of discharge and DOC, one in a clear-cut and other in relatively mature forest. Hourly data of DOC, from UV sensor, and stream discharges and rainfall intensity, from a close meteorological station, were acquired during the winter of 2018. The DOC response to storms was larger and faster at the clear-cut site and elevated storm responses may be due to changes in flow paths related to forest harvesting. However, low antecedent flow and greater storm intensity lead to significantly elevated DOC storm response.

A review by Korshin et al. [50] examines the potential of spectroscopic methods (absorbance and fluorescence) for assessing the efficiency of water treatment. The variations of absorbance and fluorescence values along a treatment are linked to the reduction of dissolved organic pollution and also correlated with the formation of by-products. The review concludes that the current level of the development and implementation of spectroscopic methods for online/real-time water quality monitoring is far from its real potential.

A recent application by Sahraei et al. [51] described a mobile trailer for high-frequency measurements of stable isotopes and other water quality parameters (20 minutes interval) from groundwater, stream water, and precipitation. UV sensor was used for nitrate and surrogate parameters and the results explain the response of runoff components and the impact on antecedent wetness periods.

Moreover, several applications based on high-frequency monitoring of DOC and nitrate were recently proposed for understanding biogeochemical and hydrological processes and nutrients dynamics in watersheds [52–55].

As a conclusion of this section, Blaes et al. [56] recalled the importance of optical techniques for in situ nutrient monitoring (nitrate and DOM) and reviewed several studies carried out between 2011 and 2015 using optical systems (colorimetric, UV, and fluorescence probes) for

half a month to 4 years. Considering that unnecessarily high resolution of sampling can result in excessive memory and power consumption, and certain monitoring techniques can increase maintenance requirements and waste generation They proposed an adaptive monitoring scheme for reducing monitoring frequency during low biogeochemical activity or variability (during low flow period, for example) and increasing the measurement rate for biogeochemically active period (e.g., high flow events). This procedure can be managed from data variations monitoring of physical linked parameters (rainfall, water level, etc.) for a remote adjustment of water monitoring frequency.

Among the different techniques proposed for high-frequency monitoring of nitrate and DOM in freshwater, UV sensors are the most frequently used. For example, the results of field experiences using UV/Vis sensors for high-resolution monitoring of nitrate in groundwater concluded recently that high-resolution UV/Vis monitoring should greatly contribute to a better understanding of groundwater processes in the future [57]. Another study [58] on the use of a UV–Vis submersible spectrophotometric probe confirmed that this technique facilitates rapid and robust measurement of the DOC content under remote field conditions. It also showed that qualitative information about DOM composition are possible through the ratios of absorbance values at specific wavelengths. For longer monitoring periods, the performance of UV sensors can be limited by turbidity [57] or fouling [59], and a specific attention (local calibration, maintenance) is thus required.

A recent experimentation, using a homemade UV system with a dual pathlength [42] for high-frequency monitoring of river water during a rain period in Brittany (France), integrated into a field monitoring station with multiprobe (physicochemical parameters), flowmeter and sampler, confirmed the relevance of high-frequency monitoring [60]. In order to study the impact of rainy events on DOC evolution of water, the monitoring station was installed at the mouth of the Fremur River (Brittany, France). During 9 days, including two heavy rainfalls, UV spectra were acquired every 15 minutes and the second-derivative spectra were calculated (see Chapter 3) from spectra for DOC estimation at 295 nm [42]. Fig. 9.2 and Fig. 9.3 show, respectively, the set of second-derivative spectra and the evolution of river discharge and of the value of the second derivative at 295 nm.

Finally, high-frequency monitoring of water quality can strongly improve the understanding of the complex spatial patterns and temporal dynamics of biogeochemical cycling in systems sensitive to environmental change [56,61]. Considering that high-frequency monitoring of nutrients is subject to research and pilot studies up to now, a transition from research to implementation in operational practice is needed [62].

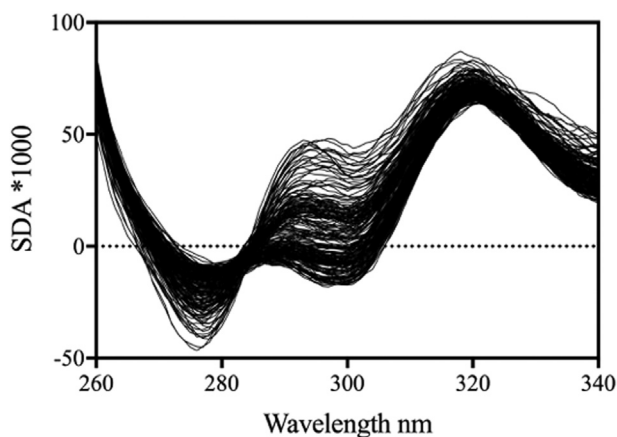


FIGURE 9.2 Set of second-derivative spectra acquired during 9 days.

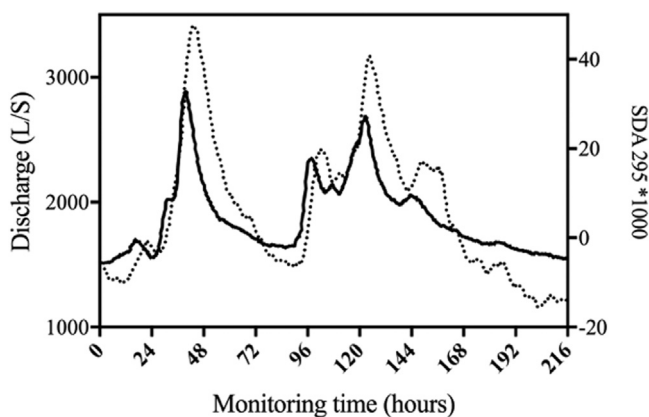


FIGURE 9.3 Evolution of discharge measured at the mouth of the Fremur river (Brittany) (*solid line*), during two rainy events, and corresponding second derivative absorbance value at 295 nm (*dotted line*).

This transition requires efficient and cost-effective monitoring programs designed from clear monitoring objectives.

9.5 Underwater applications

Autonomous underwater vehicles (AUV) were developed in the last century for scientific (marine geosciences) and military purposes [63]. From the 2000s the developments of AUV have rapidly emerged as vital

tools for marine geoscientists, especially those involved in seafloor mapping and monitoring [64].

In a recent study, de Lima et al. [65] described the outcomes, field experiences, and feedback gathered from the use of underwater drones equipped with sensors and video cameras in various pilot applications in The Netherlands, in collaboration with local water managers. In the past decades, the first experiments were the exploration of deep layers of the ocean. Numerous works dealing with reef monitoring using video cameras and underwater sampling were also possible. Then, underwater drones were equipped with water quality monitoring instruments such as camera, multiparameter probes, and algae optical sensors for turbidity, chlorophyll-a, and phycocyanin measurements. Data can be acquired every 10 seconds due to the frequency limitation of the dissolved oxygen sensor. The insights gained by using underwater drones as a mobile monitoring system showed that, in many cases, these tools are a valuable complement to other methods, as they provide a unique combination of three-dimensional data and underwater images collection.

In their paper, Pena-Pereira et al. [66] described a sensor system boarded into an AUV for real-time acquisition of water quality parameters. The AUV called CWolf from the Fraunhofer Institute was a 7' long, 1' diameter, and 135 kg weight submersible boarding a multisensors set for the in situ measurement of dissolved oxygen, conductivity, temperature, and nitrate by UV spectrophotometry. It was used for the study of the impact of fish farms on seawater quality thanks to selected dive profiles and monitoring campaigns.

9.6 Wireless sensor networks

Before completing this chapter, let us consider wireless sensor network (WSN) which represents a promising technology for water quality monitoring and management. WSN began to be used for water quality monitoring in the early 2000s but have gained increasing attention in recent years as the devices and communication techniques have improved [67].

Initially, conceived around the measurement of basic physicochemical water quality parameters (pH, temperature, dissolved oxygen, conductivity, oxydo-reduction potential, and turbidity) [68], the concept of WSN moved toward more sophisticated systems such as the one recently described by Martínez et al. [69] integrating an online ionic chromatographic analyzer with a UV detector for nitrate and nitrite measurement.

A promising project was recently proposed by Chen and Han [70] in the frame of a pilot project around the smart city infrastructure Bristol is open (United Kingdom). The sensing part is a multiparameter probe

for the measurement of pH, temperature, dissolved oxygen, conductivity, oxydo-reduction, and turbidity, with TSS estimated from the turbidity value, TDS estimated from conductivity value, and salinity calculated from conductivity and temperature. The system is completed by a video camera to capture video images of the water surface. Several systems were deployed on harbor–wharf. The water quality sonde was immersed near water surface (approximately 50 cm deep) and the camera firmly placed on ground. A data transmission module utilized the common harbor Wi-Fi network. The great interest of this system is the possible exploitation of water surface image for the detection of oil spill or algae blooms. Once the sky reflectance is treated, images captured by satellites could be also considered for a better water quality monitoring.

A long-term observation was recently reported by Ye and Kameyama [71] with the monitoring of 15 water quality parameters in Japan from 1982 to 2016. Basic physicochemical and pollution parameters, *E. coli* and some organics (chloroform, formaldehyde, nonylphenol, and anionic surfactants) were measured after sampling in the whole Japan (at 6484 sites). Several parameters require a spectrophotometric detection, often after a specific reaction. After data cleaning, it was shown that wastewater management has been effective throughout Japan, but water degradation was registered at some sites which will be the targets for future management and monitoring.

The seasonal variability of stream water quality response to storm events captured was studied by Fovet et al. [72]. A 4-year dataset covering 177 storm events and combining high-frequency records of stream flow, turbidity, nitrate and DOC concentrations, and piezometric levels was used to characterize the storm responses in a headwater agricultural catchment (Brittany, France). Among field sensors, a UV–visible portable probe was deployed at the outlet of the watershed. Given the storm patterns around storm events, the role of groundwater fluctuations as a major control factor was underlined. This study demonstrated that high-frequency records of water quality are important for explaining the variability of catchment storm responses

9.7 Remote sensing techniques appraisal

Even if airborne remote sensing techniques are promising and give always numerous applications in different domains, including water quality monitoring, there exist some limitations, which are as follows:

1. The occurrence of satellite observation with the presence of an adapted sensor must be compatible with the objectives of experiment (revisit time, spatial resolution),

2. The presence of clouds during the satellite passage does not allow a relevant data acquisition and data acquired shall not be considered in this case,
3. The time to result is depending on the availability of working images, data pretreatment (e.g., atmospheric and geographical corrections), and on the existence of estimation model requiring a strong calibration from in situ sampling and analysis,
4. The empirical models used for the estimation of one parameter in one water body are site specific and cannot be generalized,
5. Due to the coarse spectral and spatial resolution of the satellite dataset, as well as methodological limitations, only spectral band ratio methods, empirical methods, and two biooptical inversion methods were employed,
6. Considering the total costs of construction of satellites/aircraft and sensors, launching, operation, and image processing, airborne remote sensing can be much expensive than other ground techniques if it is especially dedicated to water quality monitoring.

For other remote sensing techniques applied for water quality monitoring, the following limitations and perspectives can be noticed (Table 9.2).

Several advantages can be drawn from these “ground” techniques among which are the precision and rapidity of measurement and the relative low-cost deployment once the sensors/system acquired. On the other hand, the current level of the development and implementation of spectroscopic methods for online/real-time water quality monitoring is far from its real potential [50].

However, the effort to be made in order to get the same level of information that the one given by airborne remote sensing techniques must

TABLE 9.2 Nonsatellite remote sensing techniques.

Monitoring technique	Water body	Limitations
Boats	Ocean, large streams	Costs of construction and operation
Buoys, floats	Coastal, rivers, lakes	Few systems available, autonomy
On-site, online	Rivers, wastewater, drinking water	Logistic, cost of systems
Handheld, portable	Small streams, ponds, lakes	Few optical sensors, field operation costs
Underwater	Ocean, coastal	Cost, heavy deployment
Wireless networks	Coastal, rivers, lakes	Few experiences, communication network, data management, data security

be great to cover the same area of observation. They are better tools for local or high-frequency monitoring, depending on research and operational questions.

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